PROJECT ROADMAP & TECHNICAL BENCHMARK REPORT

Predictive Claims Valuation in Workers' Compensation

Authors: Marius Occhiminuti · Gaëtan Amiens | **Course:** Data Mining & Machine Learning (DMML) – W1–W10 Synthesis

1. Executive Business Context & Problem Framing

In Workers' Compensation insurance, setting accurate financial reserves early in the claim lifecycle is critical for institutional solvency and effective claims management. Initial case estimates (`InitialCaseEstimate`) set by claims adjusters are inherently subjective and prone to major revisions as medical treatments unfold.

- **Business Objective:** Build an automated, data-driven, leak-proof machine learning pipeline to accurately predict the final settlement cost (`UltimateIncurredClaimCost`).
- **Operational Value:** Enable early triage of severe claims (Top 10% highest cost) to assign experienced adjusters, optimize financial reserve allocation, and mitigate severe tail-risk losses.

2. Pipeline Architecture & Data Engineering (W1–W10 Progress)

The dataset comprises 100,000 historical claim records combining demographic attributes (`Age`, `Gender`, `MaritalStatus`), financial status (`WeeklyPay`, `HoursWorkedPerWeek`, `DaysWorkedPerWeek`), family dependents (`DependentChildren`, `DependentsOther`), temporal markers, and unstructured text (`ClaimDescription`).

- **Feature Engineering:** Domain-specific feature `ReportDelayDays` calculated strictly as (DateReported−DateTimeOfAccide clipped at zero.
- **Strict Leak-Proof Protocol:** Data is split into 80% Train (80,000 rows) and 20% Test (20,000 rows) *prior* to fitting any transformer. Scikit-learn `Pipeline` and `ColumnTransformer` wrap `RobustScaler` for numerical features and `OneHotEncoder` for categorical features.
- **NLP Feature Extraction (Week 10):** Integrated `TfidfVectorizer(max_features=300, stop_words='english', ngram_range=(1, 2))` directly into the preprocessing pipeline to capture medical severity keywords from `ClaimDescription`.
- **Interactive Environment Migration:** Converted the pipeline into a modular Jupyter Notebook (`workers_comp_explorati` ensuring single-pass RAM data loading and fast iterative experimentation.

3. Target Transformation & Prediction Safety Guardrails

The target `UltimateIncurredClaimCost` exhibits an extreme heavy-tailed, right-skewed distribution.

- **Log Transformation:** Models are trained on $\log(1+y)$ to stabilize variance and compress extreme values.
- **Log-Prediction Clipping:** Raw log predictions are clipped to $[\min(\log(1+y_{\text{train}})), \max(\log(1+y_{\text{train}}))]$ before applying $\exp(\hat{y})-1$, preventing exponential explosions on test data.

4. Empirical Benchmark Results (Full 100,000 Records Dataset)

The benchmark was executed on the full 100,000-record dataset (20% held-out test set, 20,000 claims). Results are summarized below:

Model Family	RMSE ($)	MAE ($)	MAPE (%)	R^2 Score	Fit Time (s)
`random_forest`	**48,847.89**	10,567.92	265.89	**+0.0190**	109.64
`lightgbm`	48,864.09	**10,551.61**	**262.45**	+0.0184	**8.33**
`gradient_boosting`	48,899.37	10,555.83	261.67	+0.0170	106.21
`dummy_mean_baseline`	50,442.06	11,189.54	313.25	-0.0460	2.98
`lasso_regression`	77,954.11	12,292.03	514.92	-1.4983	4.34
`linear_regression`	96,926.61	13,627.84	487.97	-2.8623	5.01
`ridge_regression`	97,429.99	13,660.14	488.16	-2.9026	4.61

5. Key Empirical Conclusions & Technical Takeaways

- **LightGBM Efficiency & Accuracy Dominance:** LightGBM achieves the best absolute error (MAE = $\mathbf{\$10,551.61}$), saving $638 per claim compared to the dummy baseline, while training in just **8.33 seconds** (13× faster than Random Forest and Gradient Boosting).
- **Complete Breakdown of Linear Models:** Linear, Ridge, and Lasso regressions fail completely under the heavy-tailed target distribution (MAE > $12,290, negative $R^2 < -1.49$), demonstrating that claim valuation relies heavily on non-linear feature interactions.
- **Validation of Tree Ensembles:** Tree-based architectures capture non-linear relationships, yielding positive R^2 scores and stable predictions across all metrics.

6. Final Submission Action Plan & Roadmap (Weeks 10–12)

To maximize predictive power and prepare the final 6–8 page report, the following milestones will be executed:

1. **Milestone 1 (TF-IDF Feature Inspection):** Extract and analyze top TF-IDF n-gram feature importances (e.g., *"fracture"*, *"surgery"*, *"lower back"*) from LightGBM to identify high-cost medical terms.

2. **Milestone 2 (Deep Embeddings with DistilBERT):** Fine-tune a pre-trained Transformer (`distilbert-base-uncased`) on `ClaimDescription` to extract dense contextual embeddings of injury severity.

3. **Milestone 3 (LightGBM Hyperparameter Tuning & Multimodal Fusion):** Combine tabular and text embeddings into a unified LightGBM / Neural architecture; optimize `learning_rate`, `num_leaves`, and `colsample_bytree` via Optuna cross-validation.

4. **Milestone 4 (Model Interpretability via SHAP):** Generate global SHAP summary plots and local SHAP force plots to provide claim adjusters with clear feature attribution for high-risk claims.

5. **Milestone 5 (Final Report & Code Consolidation):** Draft the 6–8 page LaTeX final report and finalize the GitHub repository documentation.